

Technology Stack

Date	04 JULY 2026
Team ID	XXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript	Used to create a responsive, user-friendly web interface for entering soil parameters and displaying crop recommendations.

2	Backend / Server-Side	Python, Flask	Flask handles HTTP requests, processes user input, loads the trained Machine Learning model, and returns crop prediction results efficiently.
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3	Database / Data Storage	CSV Dataset (Crop Recommendation Dataset)	The agricultural dataset is stored in CSV format and is used for training and testing the Machine Learning model. <i>(If your project does not use MySQL or SQLite, this is the correct answer.)</i>
4	Cloud / Hosting / Deployment	Local Host (Flask Development Server)	The application is developed and tested locally using the Flask server. It can later be deployed to cloud platforms such as Render, Railway, or PythonAnywhere.

5	Version Control & CI/CD	Git, GitHub	Git is used for version control, while GitHub stores the project repository and supports collaboration and future deployment.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Joblib, Jupyter Notebook, VS Code	Scikit-learn is used to build the Machine Learning model, Pandas and NumPy are used for data preprocessing and analysis, Joblib stores the trained model, Jupyter Notebook is used for experimentation, and VS Code is used for application development.